



Optimization of N doping in TiO₂ nanotubes for the enhanced solar light mediated photocatalytic H₂ production and dye degradation[☆]



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ABSTRACT

Herein, we report the optimization of nitrogen (N) doping in TiO₂ nanotubes to achieve the enhanced photocatalytic efficiencies in degradation of dye and H₂ gas evolution under solar light exposure. TiO₂ nanotubes have been produced via hydrothermal process and N doping has been tuned by varying the concentration of urea, being the source for N, by solid-state dispersion process. The structural analysis using XRD showed the characteristic occupancy of N into the structure of TiO₂ and the XPS studies showed the existence of Ti–N–Ti network in the N-doped TiO₂ nanotubes. The obtained TEM images showed the formation of 1D tube-like structure of TiO₂. Diffuse reflectance UV–Vis absorption spectra demonstrated that the N-doped TiO₂ nanotubes can efficiently absorb the photons of UV–Vis light of the solar light. The optimized N-doped TiO₂ nanotubes (TiO₂ nanotubes vs urea @ 1:1 ratio) showed the highest degradation efficiency over methyl orange dye (~91% in 90 min) and showed the highest rate of H₂ evolution (~19,848 μmol h⁻¹g⁻¹) under solar light irradiation. Further, the recyclability studies indicated the excellent stability of the photocatalyst for the durable use in both the photocatalytic processes. The observed efficiency was ascribed to the optimized doping of N-atoms into the lattices of TiO₂, which enhanced the optical properties by forming new energy levels of N atoms near the valence band maximum of TiO₂, thereby increased the overall charge separation and recombination resistance in the system. The improved reusability of photocatalyst is attributed to the doping-induced structural stability in N-doped TiO₂. From the observed results, it has been recognized that the established strategy could be promising for synthesizing N-doped TiO₂ nanotubes with favorable structural, optical and photocatalytic properties towards dye degradation and hydrogen production applications.

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1. Introduction

Environmental contamination and energy scarcity are the major societal challenges of the human needs. A broad variety of processes and nanomaterials have been explored for sustainable

remediation of environmental pollution and the eco-friendly techniques such as photocatalytic and electrocatalytic process has acquired the top-priority for the decomposition of toxic organic dyes and storable chemical fuel production namely the hydrogen (H₂) generation (Theerthagiri et al., 2014, 2020a, 2020b; Bai et al., 2019; Reddy et al., 2020; Shwetharani et al., 2020; Wang et al., 2020). In this direction, titania based nanotubes (TNTs) have attracted considerable interest thanks to their unique features like one dimensional (1-D) morphology, high surface area, good

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adsorption ability and unidirectional transfer of charge carriers along the axis of the tubular morphology (Lakshmana Reddy et al., 2019; Lee et al., 2014; Zhao et al., 2014).

TiO₂ nanotubes displayed potential applications in diverse fields including photocatalysis for pollutants degradation and H₂ generation, dye-sensitized solar cells, photochromic devices and gas sensors (Asahi et al., 2001; Chen and Mao, 2007; Lakshmana Reddy et al., 2018a, 2018b; Martha et al., 2012; Pan et al., 2011; Tian et al., 2014). However, the key limitations of utilizing TiO₂ nanotubes are their relative wide bandgap (3.0–3.2 eV) and rapid recombination rate of photo-induced electron–hole pairs (Hou et al., 2013; Kment et al., 2017; Lakshmana Reddy et al., 2017a, 2017b). To overcome these drawbacks, a number of strategies such as doping of transition metal cations (Reddy et al., 2011; Sun et al., 2012; Jayaraman et al., 2015; Theerthagiri et al., 2015, 2020a, 2020b; Fujita et al., 2016; Wu et al., 2014) and nonmetals (Table 1, 2 and 3; Ansari et al., 2016; Basavarajappa et al., 2020; Ma et al., 2014; Sudhaik et al., 2018; Wang et al., 2013), dye sensitization (Du et al., 2012; Park et al., 2016) and modification of surface by noble metal decorations and composite formation with low band gap semiconductors (Kumar et al., 2016; Lian et al., 2015; Manchala et al., 2019; Lakshmana Reddy et al., 2017a, 2017b; Rosseler et al., 2010; Theerthagiri et al., 2019) have been reported. Among various dopants, the non-metal dopants (N, P, S) have received great attention owing to their built-in properties similar to the oxygen atom. Hence, the doping of non-metals into the lattices of TiO₂ nanotubes is one of the most important and potential methods to increase the photocatalytic activity.

Among non-metals, the nitrogen gains substantial attention because doping of N atoms into the TiO₂ lattice structure leads to the generation of N 2p energy levels near to the valence band, thus causing the reduction of bandgap in TiO₂ and the absorption of light is extended to the visible spectrum (Asahi et al., 2014). Further, the N atoms also prevent the recombination of charge carriers in N-doped TiO₂ (Asahi et al., 2014). Besides, the N atoms are able to form metastable centers because of its stability, low ionization potential and atomic radius comparable with oxygen (Devi and Kavitha, 2013). Because of the potential applications of N-doped TiO₂ nanotubes in various fields, a wide variety of synthetic routes like solid state dispersion method (Liu et al., 2014), ion-implantation method (Kalantari et al., 2016), chemical bath deposition and sol-gel (Mahy et al., 2018; Wang et al., 2014), ammonia assisted annealing (Asahi et al., 2014; Xing et al., 2009), anodization of Ti foil in nitrogen containing electrolyte precursor (Zhang et al., 2015) and other methods have been developed (Stan et al., 2019).

Among them, the solid-state dispersion method is found to be the most effective approach for doping of N into the lattice of TiO₂. Notably, this method involves a straightforward procedure of simple grinding the nano powder of TiO₂ and nitrogen precursors (urea in this work) with little amount of solvent followed by drying and calcination at relatively low temperature (Lakshmana Reddy et al., 2017a, 2017b, 2018a, 2018b). Therefore, in the current work, we have adopted this straightforward approach for the preparation of N-doped TiO₂ nanotubes for effective photocatalytic activities towards the solar light activated H₂ production and the degradation of organic dye.

2. Experimental

2.1. Materials and reagents

In the present work, the following chemicals and reagents were used without further purification: TiO₂ (anatase 99%), Urea, NaOH pellets (97%), Glycerol and HCl (0.1 M) (37%) are procured from Merck, India. Methyl orange dye is obtained from the Sigma

Aldrich, USA. Ethanol (99.9%) (C₂H₅OH) is procured from C.S.F Chemicals, China. Double distilled water is employed for the synthesis of titanate nanotubes.

2.2. Synthesis of TiO₂ nanotubes and N-doped TiO₂ nanotubes

N-doped TNTs (NTNTs) were prepared through a step-wise procedure as shown in Fig. 1. To achieve the target NTNTs, initially, TiO₂ nanotubes (TNTs) were prepared by hydrothermal method (Praveen Kumar et al., 2013) and finally NTNTs were synthesized by simple solid-state dispersion method (Wang et al., 2018).

2.2.1. Hydrothermal synthesis of TiO₂ nanotubes

2.5 g of anatase-TiO₂ nanopowder (99%, Merck) was added into a 200 mL of 10 Molar sodium hydroxide (NaOH) aqueous solution under magnetic stirring for 1 h. Thus, the obtained milk-white suspension-solution was transferred into a Teflon-coated stainless steel (SS) autoclave. After closing the vessel, the mixture was heated in a pre-heated electric oven at 130 ± 2 °C for 20 h. The temperature of autoclave vessel was naturally brought down to the room temperature (RT) and the obtained product (white precipitate) was washed repeatedly using RO water, double distilled water and HCl (0.1 M) followed by ethanol. Then the product was dried in an electric oven at 80 °C for 12 h. Finally, the obtained dried TNTs were grounded well and calcined at 350 °C for 3 h (with a heating rate of 5 °C per minute) in a muffle furnace.

2.2.2. Synthesis of N-doped TiO₂ nanotubes

The N-doped TiO₂ nanotubes (TNTs) were prepared by solid state dispersion method using TiO₂ nanotubes and the urea as nitrogen precursor. Initially, a small amount of ethanol (1.0–2.0 mL) was added to the mixture of TNTs and urea as indicated in Table 1 and grounded well in a mortar-pestle. Each mixture was transferred into different ceramic crucibles and calcined at 350 °C for 3 h (with the heating rate of 5 °C per minute) in muffle furnace. Finally, the obtained N-doped TNTs were named sequentially with increasing urea precursor (0.2, 0.4, 0.6 and 0.8 g) labelled as NTNT-1, NTNT-2, NTNT-3 and NTNT-4.

2.3. Characterizations

The TNT and NTNT photocatalyst properties were studied using different characterizations techniques. The morphology of the catalysts was imaged using the HR-TEM (FEI-Tecnaï-F20 S-Twin) operating at 200 kV. Crystal structure of TNTs and NTNTs was investigated using X-ray diffractometer (Bruker D8 Advance XRD) with Cu K α radiation (λ of 1.54 Å) source in the range of $2\theta = 20-75^\circ$. The chemical oxidation state and surface chemical-composition of elements in the catalyst were determined using X-ray photoelectron spectroscopy (ESCA LAB-250 Xi-XPS) with Al K α source. FTIR spectrum of anatase TiO₂, TNTs and NTNTs were obtained on Perkin Elmer FT-IR Spectroscopy in the range of 4000–500 cm⁻¹. DRS UV–visible (UV–Vis) spectrum of photocatalysts was obtained using Varian Cary 5000 UV–Vis–NIR Spectrophotometer. The UV–Vis absorption spectrum and optical density of dye solution were obtained using a UV–Vis double beam spectrophotometer (UV-1800, Shimadzu). The photoluminescence (PL) spectra were recorded by dispersing catalyst in acetone medium using Agilent Technologies Cary Eclipse fluorescence spectrophotometer at excitation wavelength of 320 nm.

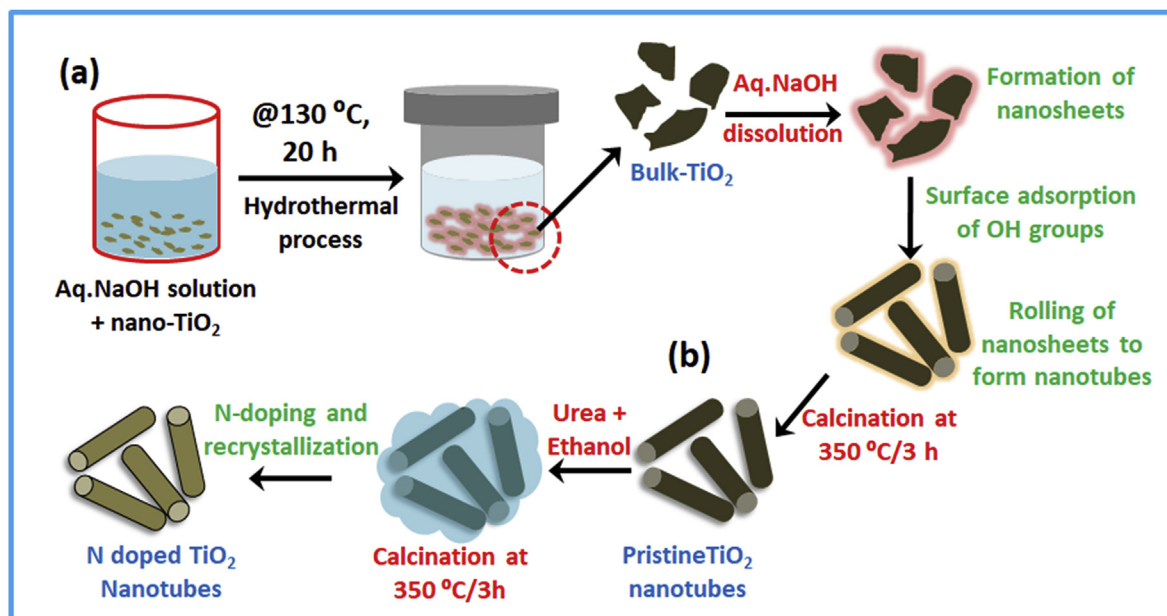


Fig. 1. Schematic depiction for the synthesis of (a) TNTs by hydrothermal method and (b) N-doped TNTs by solid state dispersion method.

Table 1

Atomic wt% of nitrogen in different NTNT catalysts.

S. No.	TNT (g)	Urea (g)	Atomic wt% of Nitrogen	Ratio	Sample ID
1	0.4	0.2	4.7	1:0.5	NTNT-1
2	0.4	0.4	9.3	1:1	NTNT-2
3	0.4	0.6	14.0	1:1.5	NTNT-3
4	0.4	0.8	18.6	1:2	NTNT-4

2.4. Photocatalytic activity experiments

2.4.1. Photocatalytic degradation of methyl orange

The photocatalytic efficacies of the TNT and NTNT were analysed using methyl orange (MO) dye as a model pollutant under solar light irradiation (average light intensity of around 1.2×10^5 lx measured by using digital lux meter (TES 1332 A)) between 12:00 and 15:00 h. The photo-catalytic studies were performed in a batch-reactor (with reactor volume of 250 mL) fabricated using borosilicate glasses. The reactor has a round-bottom shape with top-port to handle the reaction mixture. The probable evaporation of water via thermal process under light irradiation was mitigated by closing the open-end with a rubber septum. The experimental parameters were optimized based on our previous observations during the photocatalytic degradation of contaminants in a solution mixture (Lakshmana Reddy et al., 2019; Praveen Kumar et al., 2016). Before to begin the degradation experiment, 200 mg of catalyst was dispersed into a 100 mL of methyl orange (MO) aqueous solution (3×10^{-5} mol L⁻¹) at pH around 6.5 and stirred for 30 min in dark to make sure the equilibrium adsorption-desorption condition. Under the solar light exposure, at periodic intervals, around 3 mL of the dye-photocatalyst solution was taken, centrifuged and the supernatant was collected, and the optical density (OD) was assessed at 464 nm using the UV–Vis absorption spectrophotometer. The process of decomposition of MO was estimated using equation Eqn(1):

$$\text{Degradation\%} = [(C_0 - C)/C_0] \times 100 \quad (1)$$

where, C_0 and C are concentration/absorption intensity of the

resultant dye solution at zero and at a given time, respectively. The photolysis of MO solution was performed under solar light irradiation to know the effect of photocatalyst. Further, in order to create a dark condition, the glass-reactor was covered with double layered Al foil sheets. Furthermore, the reusable efficiency of the optimized photocatalyst was also tested. In this study, after collection of degraded dye solution at end of first cycle, the photocatalyst (PC) was recovered via centrifugation process, washed using distilled water and dried by keeping it in electric oven at 80 °C for 12 h. Then, this recovered photocatalyst was dispersed in a fresh MO-dye solution and the photocatalytic experiment was continued as aforementioned. This process was repeated again for four recycles and recorded results.

2.4.2. Photocatalytic hydrogen generation

Detailed experimental procedure for the solar-driven photocatalytic (PC) hydrogen (H₂) evolution is well documented in the recent reports (Kumar et al., 2016; Lakshmana Reddy et al., 2019; Praveen Kumar et al., 2016). Briefly, 0.005 g (5 mg) of the synthesized photocatalyst was dispersed into 50 mL of water-glycerol (5 vol %) mixture, which is kept in 180 mL capacity quartz PC reactor. This single port photoreactor was tightly closed with custom made rubber septum to make sure air-tight in the reactor. This set-up was then degassed and filled with high pure N₂ gas that ensured the O₂-free ambience during photocatalytic reaction under natural solar light during tropical days between 10:00 and 15:00 h (Geographical Location: 14.47°N 78.82°E Solar light intensity 1,10,000 to 1,40,000 lx). The evolved output-gases during the catalytic reaction were quantified every 1 h through an offline gas chromatograph attached with a TC detector. This equipment was optimized using the standard H₂ and O₂ gases and standard calculations were used for the quantification of the evolved output gases.

3. Results and discussion

3.1. Analysis of crystal structure

The crystal structure and phase identification of precursor

anatase TiO_2 , TNTs and N-doped TNTs were investigated by using XRD technique and displayed in Fig. 2. These patterns are well matched (JCPDS No. 21-1272) with anatase-phase TiO_2 having tetragonal crystal structure. As mentioned in the experimental part, the TNTs were originally synthesized from TiO_2 nanopowder and thereby the XRD pattern has showed significant difference in the peak shape and intensity revealing the size and morphological changes in the evolved TNTs.

Similarly, the changes have been observed in the peak shape and intensity in XRD patterns of NTNTs with respect to that of TNT. This could be attributed to the successful lattice occupancy of N and the induced reduction of crystallite size in NTNT. No drastic shift or emergence of new peaks have been observed, which indicate that the N doping has not led to formation of any secondary and impurity phases in the host TNT, rather than the more occupancy of oxygen sites in the TNT. Notably, such occupancy of N essentially modifies the electronic and optical properties of TNT intrinsically, which would expect to manifest the enhanced properties permanently in the system.

3.2. Analysis of surface functional groups

The FT-IR spectra of anatase- TiO_2 , TNT and NTNTs (NTNT-1 to NTNT-4) are shown in Fig. 3. In the spectrum, the broad band around at 3401 cm^{-1} can be ascribed to the stretching vibration of the surface-bonded $-\text{OH}$ groups (Xu et al., 2008). The weak band at 2924 cm^{-1} being observed in all the spectra corresponding to various catalysts was assigned to the stretching-vibration mode of $\text{C}-\text{H}$ bonds that could possibly from the residues produced during combustion of various precursors involved. The observed subtle peak around at 2340 cm^{-1} can be associated to the bending vibration modes of $\text{H}-\text{H}$ bonds. Peak around 1633 cm^{-1} observed in all spectra can be assigned to the bending vibration of $\text{O}-\text{H}-\text{O}$ of the physisorbed water molecules (Hu et al., 2010). Whereas, the broad band between 500 and 900 cm^{-1} was assigned to the characteristic stretching-vibration mode of $\text{Ti}-\text{O}-\text{Ti}$ bonds of anatase- TiO_2 (Hu et al., 2017; Lan et al., 2014). The peak around 1459 cm^{-1} could be corresponded to the nitrogen (N) atoms, which are substituted into lattices of TiO_2 (Geng et al., 2009), (Xu et al., 2008). Notably, this peak essentially emerges due to the $\text{N}-\text{H}$ bending mode (Matsumoto et al., 2007), indicating the appropriate doping

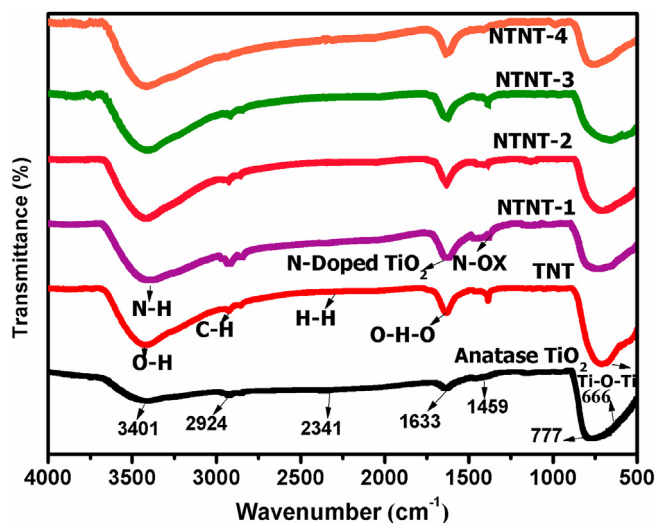


Fig. 3. FT-IR spectrum of anatase TiO_2 , TNT and NTNTs.

of N into the lattices of TiO_2 nanotubes.

Notably, the peak corresponding to $\text{O}-\text{H}$ stretching in TNTs is pronounced as compared to the bulk TiO_2 . This essentially shows the abundance existence of OH functional groups on the surface of TNTs, which could be due to their synthesis process. This OH functional groups paved ways for the further adsorption of N atoms onto the surface ($\text{N}-\text{H}$) and thereby their further diffusion into the lattices of TNTs. Similarly, the peak assigned to $\text{O}-\text{H}-\text{O}$ group in TiO_2 is changed in the case of TNTs, which could be due to the substitution of N and formation of $\text{N}-\text{OX}$ groups on the surface of NTNTs. Furthermore, the peak assigned to $\text{Ti}-\text{O}-\text{Ti}$ stretching is found to be deformed, which could be ascribed to the partial existence of $\text{Ti}-\text{N}-\text{Ti}$ network in NTNTs. These observations clearly indicate that the N has been doped into the lattices of TNTs and they possess functional groups on their surface, which could lead to the enhanced adsorption of dye molecules for their effective degradation and rapid transfer of photo-generated charge carriers towards the effective reduction of protons into molecular hydrogen.

3.3. Analysis of optical properties

The optical UV-Vis light absorption characteristics of the synthesized TNTs and NTNT-2 were investigated using diffuse reflectance UV-visible absorption spectroscopy. The obtained spectra are displayed in Fig. 4(a). The bandgap energy of the synthesized samples was estimated from Kubelka-Munk plot (Sakthivel and Kisch, 2003) as given in Fig. 4(b). It can be seen from the absorption spectra that the band edge of TNT at $\sim 405\text{ nm}$ has been shifted to $\sim 430\text{ nm}$ in the case of NTNT-2, which prepared at the 1:1 ratio of TNTs vs urea. This indicates that the N-doping has considerably expanded the optical-absorption of TNTs to visible light region. Accordingly, the bandgap energy of TNTs and NTNT-2 was measured to be 3.21 and 3.02 eV, respectively. These enhancements in the optical properties could be originated via the formation of energy levels of doped N atoms near the valence band minimum (VBM) of the host TNTs, where it fundamentally led to the dispersion of VB potential and thereby, it facilitates the reduction of band gap energy in NTNT-2 (Kim et al., 2015; Le et al., 2018).

3.4. Identification of surface morphology

The obtained electron microscopic (HRTEM) images of the

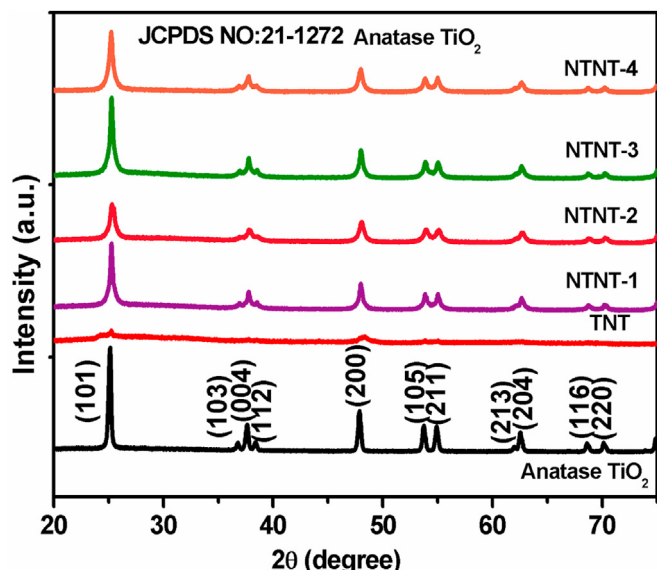


Fig. 2. XRD pattern of anatase TiO_2 , TNT and NTNTs.

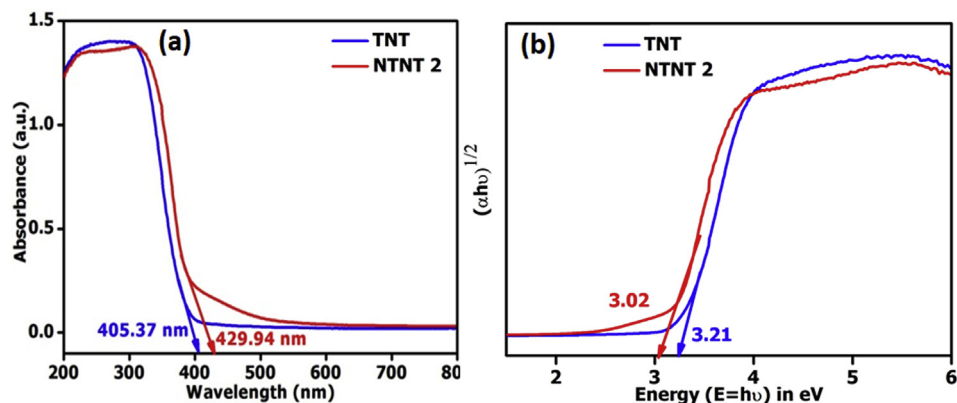


Fig. 4. (a) UV-vis absorption spectra and (b) Kubelka-Munk plots of TNT and NTNT-2.

synthesized NTNT-2 are displayed in Fig. 5(a)–(b). From the images, it is clear that the catalyst contains the mixed structure of nanotubes and particulates, where their lighter appearance is due to the small thickness of NTNT-2. However, the overall morphology of NTNT-2 is likely solid, which is completely turned into thinner due to the synthesis process mediated surface modifications in the nanotubes.

The elemental composition of NTNT-2 was investigated using HRTEM-EDX technique and the obtained spectrum showed the

presence of the respective elements (such as N, O and Ti) in NTNT-2. The observed relative intensity and concentration reveal that the stoichiometric ratio of the elements presence in the N-doped TiO_2 nanotubes. The atomic wt% of N present in each catalyst is displayed in Table 1.

3.5. Surface chemical analysis

The chemical state and surface-composition of the optimized N

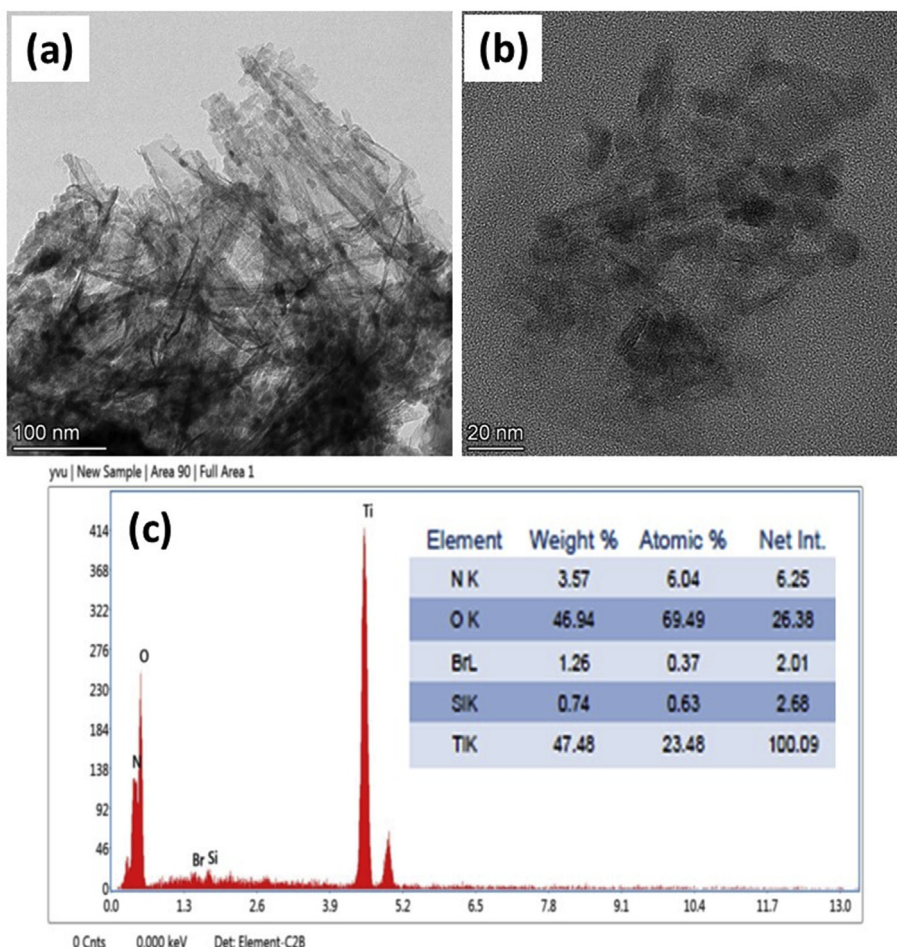


Fig. 5. (a)–(b) HR-TEM and (c) EDAX elemental spectrum of NTNT-2.

doped TiO₂ nanotubes (NTNT-2) were investigated using XPS technique. The obtained spectra are shown in Fig. 6. The full scan spectrum of the catalyst as displayed in Fig. 6(a) showed the existence of the respective elements such as N, O and Ti in the sample. The narrow scan deconvoluted Ti 2p spectrum showed (Fig. 6(b)) the major peaks at 458.2 and 464.5 eV corresponding to Ti 2p_{1/2} and Ti 2p_{3/2}, which indicated that Ti possesses +4 oxidation state (Kumar et al., 2016; Lakshmana Reddy et al., 2017a, 2017b). Notably, the existence of deconvoluted peaks and their position indicated the slight shift in the binding energy of Ti environment, which might have possibly been originated due to the existence of both Ti–O–Ti and Ti–N–Ti network in the system (Jin et al., 2019; Sun et al., 2019). Therefore, it can be realized that N has been properly doped in TiO₂ without causing any defects in material. The deconvoluted oxygen spectrum showed the peak at ~529 eV represented that the stoichiometric existence of oxygen network in TiO₂ with respect to Ti as well as the doped N (Bagwasi et al., 2013; Jia et al., 2018). Accordingly, the characteristic position of the deconvoluted peaks at the respective binding energy signified the existence of oxygen in the chemical environments of Ti–O and Ti–O–N network (Fig. 6(c)). Similarly, the deconvoluted peaks at the higher binding energy indicated the adsorbed-oxygen (O) species on the surface of NTNTs. The peak corresponding to N 1s confirmed the existence of N at the lattices of TiO₂ (Fig. 6(d)). Accordingly, the various observed peaks in the deconvoluted spectrum such as the peaks at 400.7 eV and 399.8 eV were ascribed to the substitutional nitrogen atoms in the lattices of TiO₂ and at the sites of the oxygen atoms, respectively, as in the environments of Ti–O–N and Ti–N–O in the NTNTs (Du et al., 2013; Liu et al., 2016).

3.6. Photocatalytic degradation studies

Photocatalytic degradation efficiency of the synthesized photocatalysts was evaluated on the decomposition of MO dye. The

obtained results are displayed in Fig. 7(a)–(d). From the obtained results, it was found that the optimized photocatalyst NTNT-2 (1:1 ratio of TNTs vs urea) displayed the highest degradation efficiency of ~91% in 90 min under solar light irradiation, whereas the photolysis in the absence of catalyst had no effect on the degradation of MO dye (Fig. 7(a)–(b)). It also demonstrated the best pseudo-first-order kinetics with a highest rate-constant of 0.0275 (Fig. 7(c)–(d)). The observed enhanced photocatalytic efficiency of NTNT-2 could be essentially related to the doping-induced formation of energy bands in the band gap of TiO₂. As discussed, the energy levels of N are formed near the valence band maximum and disperse the VB thereby it facilitates the effective separation of excitons (electron-hole pairs), delayed recombination and reduction of band gap energy, which absorbed more UV–visible light photons for the accelerated photocatalytic process. Furthermore, the surface adsorbed functional groups in NTNT-2, as observed from the FT-IR and XPS spectra, also contributed for the enhancements by facilitating the effective transfer of carriers to the surface and led to the rapid production of radical species for the enhanced degradations as shown in Fig. 7(e)–the inset image shows the photographic images of the degraded-MO solution collected at the periodical intervals. Further to elucidate the photocatalytic redox process on the photodegradation of MO dye, the scavenger study was carried out using ethylenediaminetetraacetic acid (EDTA) as hole (h⁺) scavenger, potassium dichromate (K₂Cr₂O₇) as electron (e[−]) scavenger and ammonium oxalate [(NH₄)₂C₂O₄] as super-oxide anion (•O₂[−]) scavenger as shown in Fig. 7(f). It is apparent that the photodegradation of MO solution is slightly inhibited by the presence of potassium dichromate or ammonium oxalate but it significantly inhibited by the addition of hole scavenger (EDTA) as compared to that of the system without addition of scavenger. This indicates that the active holes are the major reactive radicals, while the •OH and •O₂ also play key roles in the photodegradation process of MO solution (Zhang et al., 2020).

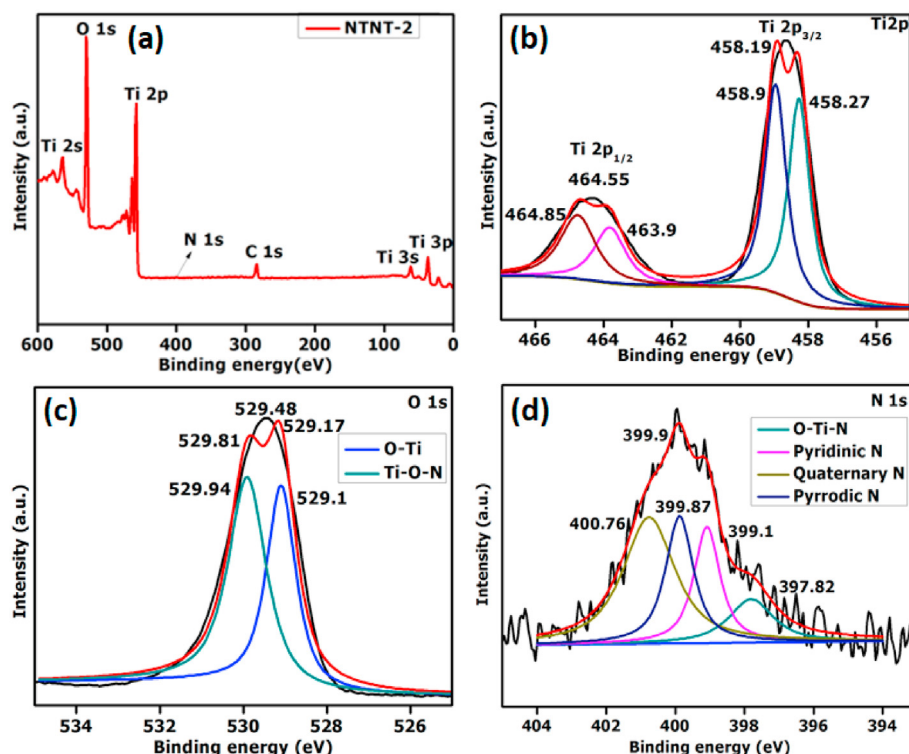


Fig. 6. XPS spectra of NTNT-2 photocatalyst (a) Survey scan, (b) Ti 2p, (c) O 1s, (d) N 1s.

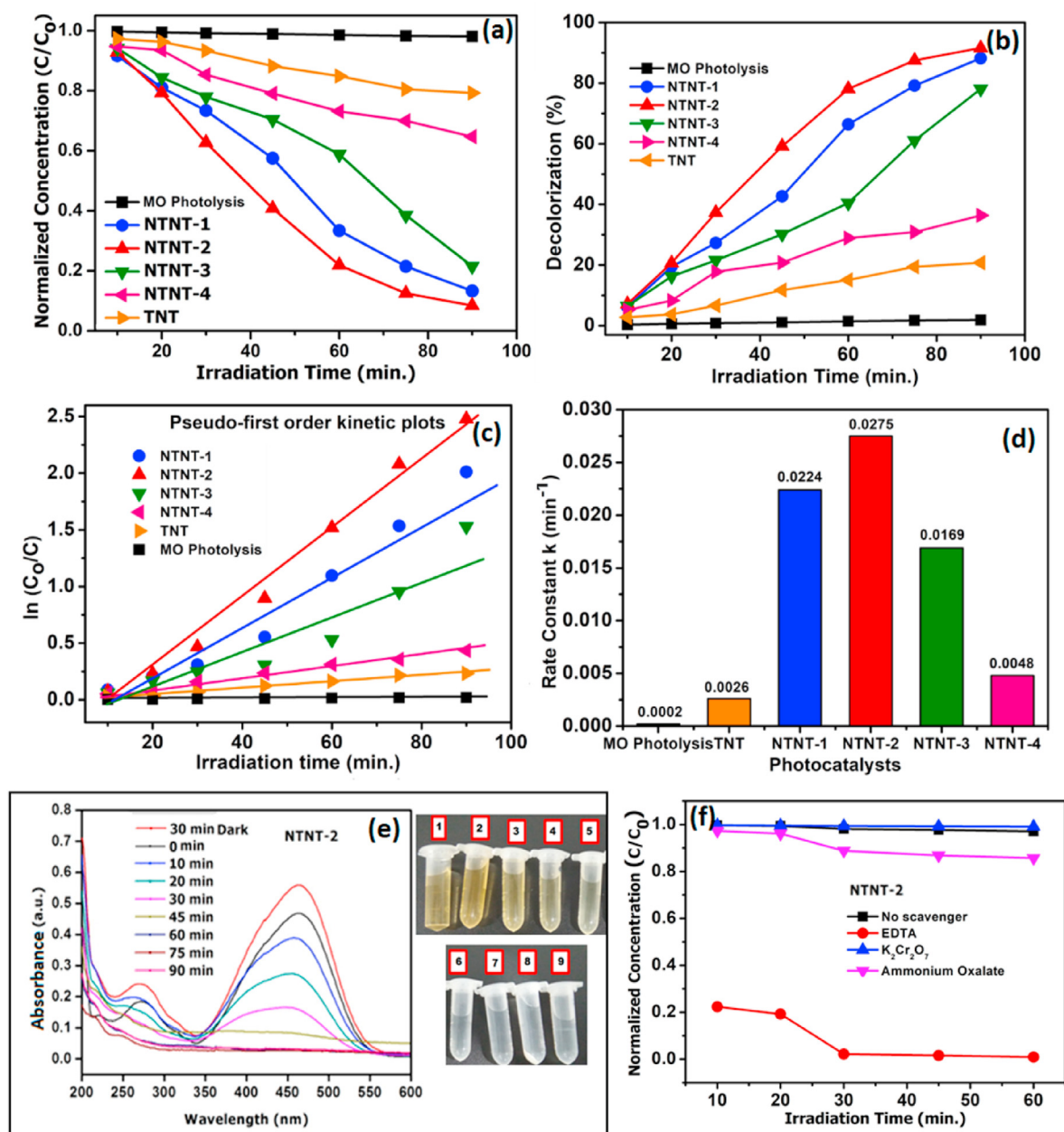


Fig. 7. Photocatalytic decolorization activities of TNT and NTNTs (a) Normalized concentrations, (b) Decolorization efficiency, (c) Pseudo-first order kinetics, (d) Rate constant determinations, (e) Variation in the absorption of NTNT-2 with photocatalytic irradiation time and corresponding photographic images of MO dye solution (inset figure) and (f) scavenger test.

Towards assessing the chemical-stability and reusable efficiency of the developed photocatalysts, the degradation experiment was repeated for four recycles for five consecutive days under identical conditions under natural solar light and the observed results are displayed in Fig. 8(a)-(b). From the results, hardly any decline in the photocatalytic efficiency was observed, which represented its excellent stability that originated due to the intrinsic properties, rich-surface activity and stoichiometric composition of the photocatalyst as observed from their XPS results.

3.7. Photocatalytic H_2 generation

The photocatalytic H_2 production ability of all the developed photocatalysts (TNT and NTNT-series) was also studied under

natural solar light exposure. The results obtained are displayed in Fig. 9(a). From the studies, the NTNT-series showed enhanced H_2 production efficiency as compared to TNT and, among the NTNT-series, the sample NTNT-2 (1:1 ratio of TNT vs urea) demonstrated the highest H_2 evolution rate of ($19848 \mu\text{mol h}^{-1} \text{g}^{-1}_{\text{cat}}$), which is around 50 folds higher as compared to that of the TNT ($390 \mu\text{mol h}^{-1} \text{g}^{-1}_{\text{cat}}$).

It should be noted that in both the photocatalytic processes such as pollutant degradation and hydrogen production, the NTNT-2 showed the enhanced activity as compared to the other NTNT-series. This could be possibly because of the N doping at lower and higher concentration with respect to the optimized concentration caused an unfavorable dynamic in the host TNT system. For instance, at the lower ratio (TNT vs urea at 1:0.5), the concentration

of N may be well below the threshold, which may not be enough to substantially lead to any changes in the properties of TNTs. Similarly, at higher ratios (TNT vs urea at 1:1.5 and 1:2), the increased concentrations of N may have screened the properties of TNTs via their occupancy at the interstitial positions of TiO_2 lattices and formation of recombination centers in TiO_2 , which declined the photocatalytic efficiency in NTNTs.

$$\text{Solar to hydrogen energy conversion efficiency (\%)} = \frac{\text{output energy (H}_2\text{)}}{\text{energy of incident solar light}} \times 100 \quad (2)$$

Interestingly, the H_2 production efficiency of the optimized photocatalyst (NTNT-2) was also investigated over various scavengers such as butanol, methanol, ethanol and glycerol in addition to water. The study indicated that the rate of hydrogen production by NTNT-2 from these sources was to be in the order of glycerol $\gg$ ethanol $\gg$ methanol $>$ butanol $>$ water. (Fig. 9(b)). The observed results are also found to be well matched with the other published reports, where the carbohydrates and alcohols were used as hole scavengers for the photocatalytic H_2 generation, such that the number of OH groups linked to the scavenger will greatly affect the oxidation reaction in the solution (Lakshmana Reddy et al.,

2018a; Lakshmana Reddy et al., 2018b; Praveen Kumar et al., 2017; N. Lakshmana Reddy et al., 2017a; N. Lakshmana Reddy et al., 2017b; Reddy et al., 2019). Further, the solar to hydrogen energy (SHE) conversion efficiency for the pristine and optimized materials has been calculated using the formula as shown in Eqn. (2).

The results of the calculations are given in Table 2. From the calculations, the TNT showed only 0.17% SHE efficiency, whereas the optimized catalyst (NTNT-2) displayed around 9.01% efficiency. It is worthy to mention that the optimized catalyst exhibited 53 folds higher efficiency than the TNT. Hence, the N-doping gradually improved the absorption of large number of photons and effectively utilized for the photocatalytic reactions towards the improved production of H_2 .

In addition, the stability of the optimized photocatalyst towards stable production of hydrogen was also investigated for four cycles using glycerol as source for protons. The obtained results are given in Fig. 9(c). The study showed that the photocatalyst tend to have consistent stability during its reuse for hydrogen production from glycerol. However, there is an increment in 2nd, 3rd and 4th cycle, which could be due to the formation of secondary products or intermediates of glycerol in the reaction solution that resulted in the increased hydrogen productions. The improved activity may also be due to the surface reaction of the OH groups that formed as intermediates, where they effectively react with the glycerol with water and making the TNT surface more of photo-catalytically active. Those active sites/functional groups helped in improved activity during the further testing cycles. Furthermore, the charge carrier transportation properties were also evaluated by PL spectroscopy and the obtained results are displayed in Fig. 9(d). From this study, it was noticed that the TNT exhibited higher PL intensity, which indicated that the electron-hole recombination rate could be high in TNT, while, the N-doped TNTs showed slightly reduced PL intensity, which revealed that the charge carrier separation has been improved in NTNTs (Lakshmana Reddy et al., 2018a; 2018b, 2019). Interestingly, the optimized catalyst (NTNT-2) showed relatively low PL intensity as compared to the TNT and other NTNTs, revealing the existence of reduced charge carrier recombination and thus it led to the generation of the highest H_2 gas than on the other NTNTs.

Finally, the observed photocatalytic efficiencies towards dye degradation and H_2 production reported in the present work are compared with the literature. Most of the literature demonstrated either dye degradation or H_2 production using the reported material and the multifunctional application of N-doped TiO_2 is very limited as shown in Table 3. From the Table; it is clear that the present study displayed relatively the better photocatalytic dye degradation and the highest H_2 production rates as compared to the literature reports. These obtained results essentially demonstrate that the systematic studies on the optimization of N-doping, the morphology control of TiO_2 and the synthesis method can drastically influence the photocatalytic activities of N-doped TiO_2 nanotubes.

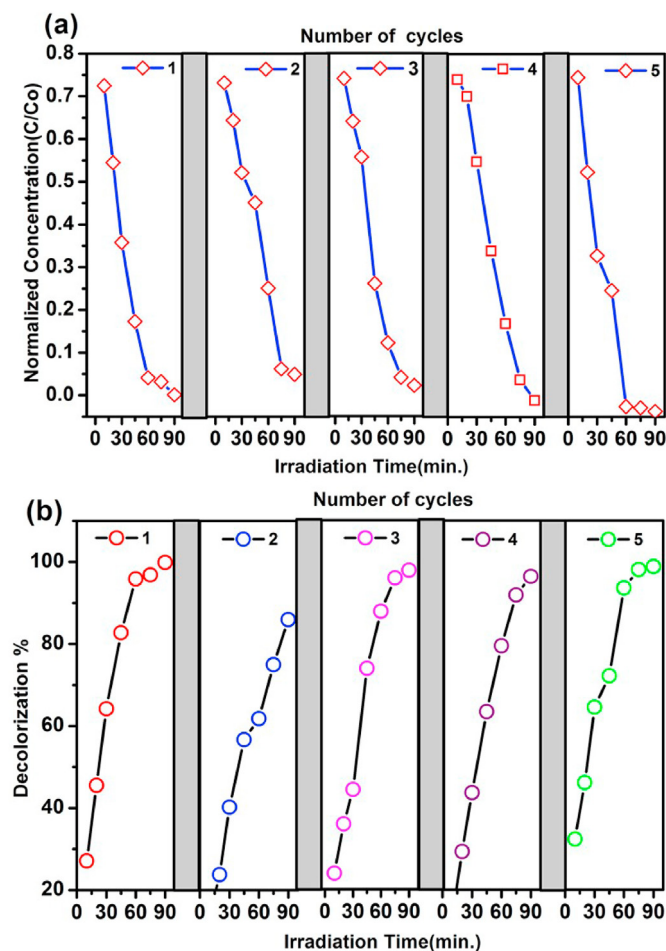


Fig. 8. Photocatalytic stability of NTNT-2 (a) normalized concentration, (b) decolorization efficiency.

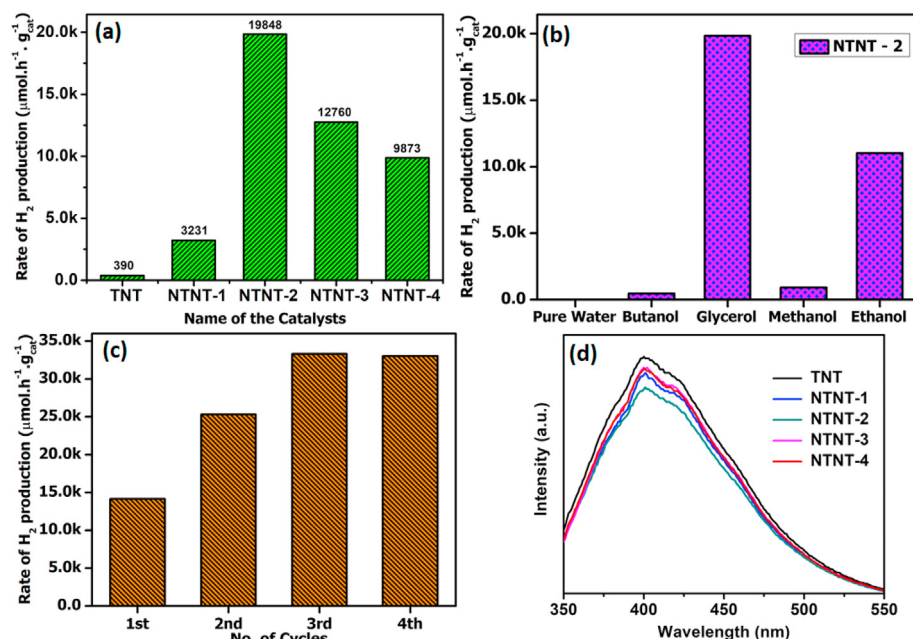


Fig. 9. (a) Optimization of hydrogen production using N-doped TNTs, and comparison with pristine TNT (b) effect of different sacrificial agents with NTNT-2, (c) Photocatalytic stability of NTNT-2 in the presence of glycerol and (d) PL spectra of pristine and NTNT samples at excitation wavelength of 320 nm.

Table 2

Solar to hydrogen energy conversion efficiency of TNT and NTNT-2 catalysts.

S. No.	Sample ID	Amount of H ₂ produced (mmol.h ⁻¹ .g _{cat} ⁻¹)	Energy of incident solar light (mW/cm ²)	SHE (%)
1	TNT	0.39	220	0.17
2	NTNT-2	19.84	220	9.01

Table 3

Comparison of photocatalytic activity of N-doped TiO₂ towards dye degradation and H₂ production.

Photocatalyst	Pollutants	Degradation activity	Hydrogen production (mmol.h ⁻¹ .g _{cat} ⁻¹)	Ref.
N-doped TiO ₂	Methylene blue & Methyl orange	92% in 60 min	—	Yang et al. (2010)
N-doped TiO ₂	Methylene blue	52% in 210 min	—	Vaiano et al. (2015)
N-doped TiO ₂ nanotubes	Methylene blue	52% in 150 min	—	Siuzdak et al. (2015)
N-doped TiO ₂ nanorods	Methylene blue	99% in 60 min	—	Zhou et al. (2014)
N-doped TiO ₂ nanotubes	Methylene blue	98% in 90 min	—	Liu et al. (2011)
N-doped TiO ₂ /C	Methylene blue	77% in 50 min	2.83	Yao et al. (2019)
N-doped TiO ₂	Methylene blue	65% in 240 min	0.02	Babu et al. (2012)
N-doped TiO ₂ nanobelts	—	—	0.67	Sun et al. (2016)
N-doped TiO ₂	—	—	0.60	Wang et al. (2013)
N-doped TiO ₂ nanotubes	Methyl orange	91% in 90 min	19.84	Present work

4. Conclusions

Nitrogen (N) doped TiO₂ nanotubes were prepared using hydrothermal method followed by simple solid-state dispersion process. The systematic characterizations of the pristine and N-doped TNT photocatalysts revealed the improved crystal structure and visible light absorption properties. The photocatalytic activity towards decomposition of methyl orange (MO) dye indicated the enhanced and stable degradation efficiency of photocatalyst under solar light irradiation, which showed around 91% degradation in 90 min. Further, the photocatalytic H₂ generation by the optimized photocatalyst was observed to be 19,848 μmol h⁻¹.g_{cat}⁻¹, which was around 50 folds higher as compared to that of the TNTs (390 μmol h⁻¹.g_{cat}⁻¹). The optimized NTNTs also showed a consistent efficiency during the recycle studies in both degradation and H₂ production process. The observed efficiency and chemical/

photostability of the photocatalyst was attributed to the optimal concentration of N-doped in TiO₂, which fundamentally induced the favorable optical, structural and surface-active properties in TiO₂ nanotubes.

Authorship statement

Venkata Divyasri Yadala: Conception and design of study, acquisition of data, analysis and/or interpretation of data, Drafting the manuscript, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Lakshmana Reddy Nagappagari:** Conception and design of study, analysis and/or interpretation of data, Drafting the manuscript, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Kiyoung Lee:** analysis and/or interpretation of data, revising the manuscript critically for important intellectual

content, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Sakar Mohan:** analysis and/or interpretation of data, Drafting the manuscript, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Navakoteswara Rao Vempuluru:** acquisition of data, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Venkatramu Vemula:** analysis and/or interpretation of data, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Shankar Muthukonda Venkatakrishnan:** Conception and design of study, acquisition of data, analysis and/or interpretation of data, Drafting the manuscript, revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published (the names of all authors must be listed). **Chinna Gangi Reddy Nallagondur:** Conception and design of study, acquisition of data, analysis and/or interpretation of data, Drafting the manuscript, revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published (the names of all authors must be listed).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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